

VOZNESENS'K, Ukraine

WMO: 337770

Lat: 47.5825N

Lon: 31.3200E

Elev: 34

StdP: 100.92

Time Zone: 2.00 (EUE)

Period: 90-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-12.0	-18.6	0.7	-14.0	-16.0	0.9	-11.0	7.7	0.3	6.7	1.0	1.8	330	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	33.7	20.5	31.7	19.7	30.2	19.2	21.7	29.6	21.0	28.7	20.2	27.7	2.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.2	24.2	18.5	13.4	23.6	17.6	12.7	23.2	63.7	29.5	60.7	28.7	58.1	27.7	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.2	4.4	-17.8	36.5	3.1	1.9	-20.0	37.8	-21.8	38.9	-23.5	39.9	-25.8	41.3		
(5)			-18.1	24.0	3.3	1.1	-20.4	24.7	-22.4	25.4	-24.2	26.0	-26.6	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-1.1	0.1	5.0	11.2	16.4	19.8	22.4	22.2	17.1	10.4	4.5	-0.3		
	DBStd	9.51	5.43	5.22	4.34	3.77	3.37	3.07	2.83	3.42	4.13	4.84	5.31	4.56		
	HDD10.0	1365	344	277	162	30	2	0	0	0	2	54	176	319		
	HDD18.3	3157	602	510	412	214	79	19	3	7	69	248	416	577		
	CDD10.0	1622	0	1	8	67	199	293	383	378	216	67	9	0		
	CDD18.3	372	0	0	0	1	18	62	128	126	33	3	0	0		
	CDH23.3	4022	0	0	0	17	202	575	1365	1483	365	16	0	0		
	CDH26.7	1517	0	0	0	2	35	173	515	670	122	0	0	0		
(14) Wind	WSAvg	2.0	2.2	2.3	2.5	2.2	1.9	1.8	1.7	1.6	1.7	1.8	2.0	2.2		
(15) Precipitation	PrecAvg	489	32	28	28	31	51	80	59	38	41	38	30	39		
	PrecMax	841	85	79	61	71	113	207	159	90	123	135	72	83		
	PrecMin	301	9	4	2	5	3	21	11	7	0	2	3	3		
	PrecStd	120	23	18	16	18	34	46	38	27	34	29	19	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	15.0	20.2	25.1	29.2	32.0	35.0	36.0	33.8	25.4	16.9	11.5		
		MCWB	7.0	9.5	11.1	16.1	17.5	20.5	22.1	20.2	20.7	17.0	13.4	9.2		
	2%	DB	7.5	10.5	16.5	22.9	27.7	30.4	32.7	34.4	29.8	22.7	14.9	9.0		
		MCWB	5.6	7.3	9.8	13.3	17.2	19.6	20.2	20.3	18.5	15.7	12.2	6.9		
	5%	DB	6.0	8.1	14.0	20.6	25.7	28.9	31.1	32.5	27.4	19.8	13.2	6.9		
		MCWB	4.6	5.6	8.3	12.4	16.6	19.5	19.8	19.9	17.5	15.0	11.4	5.7		
	10%	DB	4.9	6.3	11.7	18.6	23.9	26.8	29.6	30.6	24.9	17.7	11.5	5.2		
		MCWB	3.7	4.4	7.0	11.6	15.4	18.7	19.3	19.0	16.7	13.7	9.8	4.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	9.9	12.1	16.9	19.7	22.5	23.4	22.5	21.3	19.0	14.4	9.5		
		MCDB	8.9	13.7	18.2	25.5	25.5	29.6	31.9	31.8	29.9	22.4	16.6	10.9		
	2%	WB	6.0	7.7	10.4	14.6	18.7	21.2	21.9	21.2	19.9	17.0	12.9	7.1		
		MCDB	7.0	10.3	15.7	20.4	24.6	27.6	29.7	30.7	27.8	21.0	14.4	8.7		
	5%	WB	4.8	5.9	9.0	13.3	17.8	20.4	21.1	20.4	18.7	15.6	11.5	5.7		
		MCDB	6.0	7.6	13.0	19.1	23.5	26.6	28.6	29.8	25.1	19.0	13.0	6.8		
	10%	WB	3.8	4.7	7.6	12.2	16.7	19.4	20.2	19.7	17.4	14.3	10.0	4.4		
		MCDB	4.8	6.3	10.9	17.5	22.0	25.1	27.2	28.7	23.1	17.1	11.3	5.3		
(35) Mean Daily Temperature Range	MDBR		5.3	6.7	8.4	10.3	11.6	11.0	12.3	13.1	11.7	9.6	6.2	4.8		
	5% DB	MCDBR	6.3	9.3	12.8	14.1	14.3	13.9	14.5	16.3	15.3	12.9	8.1	6.9		
		MCWBR	5.0	6.0	7.4	6.6	5.7	5.1	4.9	5.3	6.2	7.3	5.6	5.4		
	5% WB	MCDBR	6.0	8.5	11.6	12.5	11.6	11.9	12.9	14.3	13.3	10.3	7.2	6.4		
		MCWBR	4.9	5.7	7.1	6.5	5.4	5.2	5.1	5.3	6.1	6.3	5.5	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.362	0.393	0.447	0.420	0.416	0.446	0.442	0.411	0.376	0.356	0.333		
	taud		2.259	2.208	2.213	2.093	2.255	2.290	2.215	2.219	2.271	2.351	2.332	2.315		
	Ebn at Noon		722	778	816	803	845	850	816	801	789	759	691	674		
	Edh at Noon		79	103	120	149	131	128	136	130	113	89	74	68		
(44) All-Sky Solar Radiation	RadAvg		1.01	1.83	3.06	4.59	5.86	6.27	6.19	5.55	3.85	2.38	1.09	0.78		
	RadStd		0.17	0.29	0.36	0.41	0.50	0.41	0.37	0.31	0.50	0.29	0.19	0.16		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (9 neighbors)	+0.69	N/A	N/A	+0.93	N/A	N/A	-110	-160	+120	+73

Nomenclature: See separate page